

Microeconomics Theory - Review Questions

1. The inverse demand function for apples is defined by the equation $p = 214 - 5q$, where q is the number of units sold. The inverse supply function is defined by $p = 7 + 4q$. Because of an influx of apple-loving immigrants, the demand for apples increased to $p = 259 - 5q$ causing an increase in its price. The local government decides to fight the increase in the price of apples by controlling the price at its original level.
 - a. Calculate the equilibrium price and quantity in the apples market before and after the increase in demand.
 - b. Calculate the size of the shortage that the price control policy will generate.
 - c. Calculate the change in the consumer's surplus due to the price control.
 - d. Calculate the change in the producer's surplus due to the price control.
 - e. Calculate the dead weight loss due to the price control.

2. On August 29, 2005, Hurricane Katrina caused severe damage to oil installations in the Gulf of Mexico. This damage resulted in a substantial reduction in the short run supply of gasoline in the US. Georgia governor Sonny Perdue suspended his state's gas taxes for one month (both quantity and sales tax) because "it is absolutely wrong for the state to reap a tax windfall in this time of urgency and tragedy". Other states considered taking similar measures. This question analyzes this policy.

Assume that the daily short-run supply of gasoline was perfectly inelastic at 180 million gallons before the storm and perfectly inelastic at 150 million gallons after the storm. Suppose that the demand function, measured in millions of gallons per day, is given by $Q = 240 - 30P$ where P is the dollar price, including tax, that consumers pay for gasoline.

- a. What is the market price for gasoline before and after the hurricane?
- b. Suppose that both before and after the hurricane, a government tax of 10 cents is charged for every gallon of gasoline sold in the US. How much would suppliers receive per gallon of gasoline before and after the hurricane?
- c. Suppose that after the hurricane, the government removed the gas tax. How much revenue would the government lose per day by removing the tax? Will consumers benefit? How about producers?

Suppose now that after the hurricane, the 10 cent tax is removed in some states but not others. The states where the tax was removed constitute just half of the demand in the US. Thus the demand schedule in each half of the country is $Q = 120 - 15P$ where P is the price paid by consumers in that part of the country. Let P^* be the equilibrium price for consumers in the part of the country where the tax is removed. In equilibrium, suppliers must receive the same price per gallon in all parts of the country.

- d. What is the price paid by consumers in the states that removed the tax and in states that did not remove the tax? What is the price received by suppliers?
- e. By how much will the consumption of gasoline change in states that removed the tax and in states that did not remove the tax?
- f. How much tax revenues will governments of states that remove the tax lose per day? How about governments of states that did not remove the tax?

3. Consider a market with the following linear demand and supply functions:

$$Q_d = 200 - 4P$$

$$Q_s = -40 + 2P$$

The government imposes an **ad-valorem sales tax** of $t = 20\%$ (i.e., the tax is 20% of the price). The tax is collected from producers.

- a. Calculate the equilibrium price and quantity before the tax is imposed.
- b. Calculate the new equilibrium quantity after the tax is imposed. What is the price paid by consumers? What is the price received by producers?
- c. Calculate the change in consumer surplus due to the tax.
- d. Calculate the change in producer surplus due to the tax.
- e. Calculate the government revenue generated by the tax.
- f. Calculate the dead weight loss (DWL) caused by the tax.